

Frédéric Andrian

AI Engineer

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Profile

Recent graduate of ENSC, Bordeaux INP, specializing in artificial intelligence and cognitive science. I recently completed my final-year internship at INRIA Bordeaux and am available as soon as possible. At INRIA, I developed an end-to-end AI application combining a multi-step agent, Python/FastAPI services, a RAG pipeline, MongoDB and a React/Next.js interface. I also designed its prompts and tools, evaluated several LLMs and contributed to a scientific article. I want to continue developing useful and robust AI solutions.

Experience

INRIA

Bordeaux, France

AI Research Engineering Intern

Mar. 2026 – Sept. 2026

- Independently designed and built a working AI application to help researchers find scientific papers, with guidance from my supervisors.
- Implemented a custom Python multi-step agent without an orchestration framework to understand its mechanisms, integrating the OpenAI API, tool calling, state and memory management.
- Designed prompts tailored to the research use case and tool execution, with user selection of conversation messages to include in the LLM context.
- Built a RAG pipeline with document chunking, embeddings and semantic retrieval using the ChromaDB vector database.
- Developed Python/FastAPI REST APIs documented with OpenAPI, MongoDB persistence and a React/Next.js interface for interacting with the agent and its tools.
- Automated online scientific-paper metadata collection in Python and delivered cron task results through webhooks.
- Containerized components with Docker, set up GitHub CI/CD and managed a web application using LLMs installed locally on a Mac Studio.
- Used vLLM to serve models on HPC infrastructure with NVIDIA GPUs. Compared Qwen, Gemma, GLM and quantization options using Slurm, measuring response quality, latency, TTFT and context length.
- Defined the experimental protocol, analyzed results and worked on writing a scientific article comparing language models.
- Evaluated OpenCLAW in an isolated Docker environment and analyzed risks arising from autonomous tool and file access.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

June 2025 – Sept. 2025

- Developed a reinforcement learning model and its training environment for autonomous navigation of a hospital robot.
- Deployed and optimized the software and model on a Raspberry Pi for real-world testing.

Academic Project

League of Legends Player Assistant

Python, Pandas, scikit-learn, Random Forest

- Collected match data through an API, cleaned and preprocessed it with Pandas, and trained a Random Forest model with scikit-learn to support player decisions.
- Integrated model predictions into a Discord bot to make the results accessible to players.

Skills

Development

Python, JavaScript, C#, .NET, Java, FastAPI, React, Next.js, Node.js, REST APIs, OpenAPI

Generative AI and agents

LLMs, vLLM, OpenAI API, LangChain, prompts, multi-step agents, RAG, embeddings, ChromaDB

Data and Machine Learning

Pandas, scikit-learn, SQL, MongoDB, PyTorch, TensorFlow, Deep Learning, RL

LLM evaluation

Experimental protocols, quantization, latency, TTFT, quality, context length

Software engineering tools

Linux, Docker, Git/GitHub, CI/CD, Slurm, HPC environments

Languages

French (native), English (fluent, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026

- Coursework in computer science, software engineering, AI and cognitive science, including human-computer interaction, ergonomics and user experience, with a specialization in LLMs, NLP and deep learning.